

Correction to “Rethink Repeatable Measures of Robot Performance with Statistical Query”

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Abstract

This note corrects an error in “Rethink Repeatable Measures of Robot Performance with Statistical Query” by Weng et al., *IEEE Transactions on Robotics*, vol. 42, pp. 561–578, 2026. The evaluation of the integral in equation (12) of the published article is incorrect; the error propagates into equations (13)–(15) (Theorem 2) and into the validity condition stated in Remark 4. We also surface a mild structural assumption on the conditional output law used in the CDF dominance step of Section III. The qualitative results of Weng et al. remain valid; only the closed-form constants in the accuracy/repeatability trade-off change.

1 The error in equation (12)

In Section III of [1], the probability that two outputs m_1, m_2 of an α -quantized γ -accurate algorithm fall in the same partition interval is lower-bounded under the geometric CDF dominance

$$\mathbb{P}(D \leq \delta \mid |m_i - r^*| \leq \gamma) \geq \frac{4\gamma\delta - \delta^2}{4\gamma^2}, \quad \delta \in [0, 2\gamma],$$

where $D = |m_1 - m_2|$. The argument then writes (equation (12) of [1], with δ in place of the article’s)

$$\int_0^\alpha \frac{\alpha - \delta}{\alpha} d\left(\frac{4\gamma\delta - \delta^2}{4\gamma^2}\right) = \int_0^\alpha \frac{\alpha - \delta}{\alpha} \cdot \frac{4\gamma - 2\delta}{4\gamma^2} d\delta \stackrel{?}{=} \frac{4\gamma\alpha - \alpha^2}{4\gamma^2}. \quad (1)$$

The first equality is correct; the final equality is incorrect.

2 Corrected integral evaluation

Assume $0 < \alpha \leq 2\gamma$ (the regime considered in the article). Expanding the integrand,

$$(\alpha - \delta)(4\gamma - 2\delta) = 4\alpha\gamma - (2\alpha + 4\gamma)\delta + 2\delta^2.$$

Term-by-term integration over $[0, \alpha]$ yields

$$\int_0^\alpha (\alpha - \delta)(4\gamma - 2\delta) d\delta = 4\alpha^2\gamma - \frac{1}{2}(2\alpha + 4\gamma)\alpha^2 + \frac{2}{3}\alpha^3 = 2\alpha^2\gamma - \frac{1}{3}\alpha^3.$$

Dividing by $4\alpha\gamma^2$ gives the corrected value

$$\boxed{\int_0^\alpha \frac{\alpha - \delta}{\alpha} \cdot \frac{4\gamma - 2\delta}{4\gamma^2} d\delta = \frac{6\gamma\alpha - \alpha^2}{12\gamma^2} = \frac{\alpha(6\gamma - \alpha)}{12\gamma^2}.} \quad (2)$$

3 A structural assumption for the CDF dominance step

The lower bound on the CDF of D in equation (9) of [1] coincides with the CDF of $|U_1 - U_2|$ for U_1, U_2 iid uniform on $[r^* - \gamma, r^* + \gamma]$, so it should be read under the corresponding structural assumption on the conditional law of (m_1, m_2) :

Assumption 1. *The conditional distribution of $D = |m_1 - m_2|$ given $\{|m_i - r^*| \leq \gamma, i = 1, 2\}$ satisfies $\mathbb{P}(D \leq \delta \mid \cdot) \geq (4\gamma\delta - \delta^2)/(4\gamma^2)$ for all $\delta \in [0, 2\gamma]$.*

For SQ-type sample averages, the conditional output is approximately Gaussian by the central limit theorem and is therefore more centrally concentrated than the iid-uniform reference, under which Assumption 1 is conservatively satisfied.

4 Corrected Theorem 2 and Remark 4

Under Assumption 1, substituting (2) into the chain of inequalities leading to equation (14) of [1] (i.e. $(1 - c)^2 \cdot \mathbb{P}(\text{same interval} \mid \cdot) \geq 1 - \beta$) gives the corrected form of equation (14):

$$(1 - c)^2 \frac{6\gamma\alpha - \alpha^2}{12\gamma^2} \geq 1 - \beta. \quad (3)$$

Note that (2) establishes (3) directly for $\alpha \in (0, 2\gamma]$; for $\alpha > 2\gamma$ a direct calculation gives $\int_0^{2\gamma} \frac{\alpha - \delta}{\alpha} \cdot \frac{4\gamma - 2\delta}{4\gamma^2} d\delta = 1 - \frac{2\gamma}{3\alpha}$, and one verifies $1 - \frac{2\gamma}{3\alpha} \geq \frac{6\gamma\alpha - \alpha^2}{12\gamma^2}$ for all $\alpha > 2\gamma$ because the difference equals $(\alpha - 2\gamma)^3 / (12\gamma^2\alpha) \geq 0$. Hence the polynomial bound on the right-hand side of (3) is a valid (if conservative beyond $\alpha = 2\gamma$) lower bound for all $\alpha > 0$, and rearranging as a quadratic in α and selecting the smaller root gives the corrected closed form:

Theorem 1 (Corrected Theorem 2). *Let $\mathcal{T}\mathcal{E}(\cdot)$ be a SQ-type testing algorithm with scalar output in $\mathcal{M} \subset \mathbb{R}$ that is γ -accurate (with $\gamma > 0$ and probability $1 - c$ for $c \in (0, 1)$), and suppose Assumption 1 holds. The α -quantized version of $\mathcal{T}\mathcal{E}(\cdot)$ is β -repeatable ($\beta \in (0, 1)$ and $(1 - c)^2 \geq \frac{4}{3}(1 - \beta)$) for*

$$\alpha = 3\gamma \frac{(1 - c) - \sqrt{(1 - c)^2 - \frac{4}{3}(1 - \beta)}}{1 - c}. \quad (4)$$

Remark 1 (Corrected Remark 4). *The condition $(1 - c)^2 \geq \frac{4}{3}(1 - \beta)$ is the analogue of the original $(1 - c)^2 \geq 1 - \beta$ in [1] and, together with Assumption 1, is crucial for the validity of (4). The qualitative interpretation given in the original Remark 4 is unchanged: pairwise repeatability confidence is intrinsically bounded by squared pairwise accuracy confidence, with less stable (hence less accurate) algorithms being correspondingly less repeatable. Only the constant tightens, from 1 to $\frac{4}{3}$.*

5 What is unaffected

The following results in [1] are unaffected by the error:

- Definition 1 (γ -accuracy), Definition 2 (β -repeatability), Definition 3 (α -partition), and Definition 4 (α -quantized algorithm).

- Theorem 1 (accuracy of the α -quantized algorithm: $\gamma + \alpha/2$ -accuracy with probability $1 - c$). This statement uses the value of α directly; for any given α , the accuracy bound $\gamma + \alpha/2$ continues to hold.
- The geometric setup in equations (10)–(11) of [1]: the random-offset same-interval probability $\mathbb{P}_{\text{same}}(\delta) = (\alpha - \delta)^+/\alpha$ and the same-interval expectation identity $\mathbb{P}(\text{same interval} \mid \cdot) = \mathbb{E}[(\alpha - D)^+/\alpha]$. (Equation (9), the CDF dominance, now requires Assumption 1 as discussed in Section 3.)
- All material from Section III-B onward concerning γ -accuracy of the importance-sampling estimator, the Lyapunov stability argument, and the case studies.

Only the closed-form relationship between α and (c, β, γ) in equation (15) of [1] needs to be replaced by Theorem 1 above.

6 Effect on numerical and experimental results

For any target (c, β, γ) , the corrected Theorem 1 prescribes a strictly larger α^* than the original equation (15) of [1], since the original right-hand side exceeds the corrected one by $\alpha(3\gamma - \alpha)/(6\gamma^2) > 0$ on $\alpha \in (0, 2\gamma]$. The originally claimed β -repeatability therefore remains certified by the corrected analysis whenever the α actually used in an experiment is at least α^* , i.e., whenever α was set more relaxed than equation (15). For the tighter α produced by equation (15), the certified disagreement bound weakens to

$$\beta' := 1 - (1 - c)^2 \frac{\alpha(6\gamma - \alpha)}{12\gamma^2} \geq \beta_{\text{originally claimed}}.$$

Theorem 1’s accuracy bound $\gamma + \alpha/2$ depends only on the α used and is unaffected, as are the empirical disagreement frequencies $\hat{\beta}$ reported in [1].

References

- [1] B. Weng, L. Capito, G. A. Castillo, and D. Khor, “Rethink repeatable measures of robot performance with statistical query,” *IEEE Transactions on Robotics*, vol. 42, pp. 561–578, 2026. doi: [10.1109/TRO.2025.3645934](https://doi.org/10.1109/TRO.2025.3645934).